

Further Remarks

- real-value coherence
- my "feasibility sketch"

- motivate real-value coherence as something missing from BLI
- simplifying BLI?
- hypothetical marginalization
- viability sketch

- Recall that I said there seem to be some rationality constraints missing from BLT

$$1. P_n(\phi) \approx Q_n(\phi) \text{ for small } \phi$$

$$2. P_n(\phi | Q_m = Q_m) = Q_m[\phi] \quad m > n$$

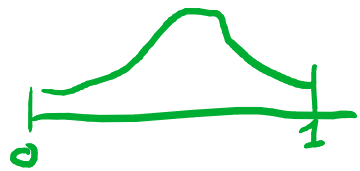
$$3. P_n(\phi | Q_m = Q_m \wedge Q_{m+1} = Q_{m+1}) = Q_{m+1}[\phi]$$

$$4. P_{n-1}(\phi) = \sum_{Q_n} P_{n-1}(Q_n = Q_n) Q_n(\phi)$$

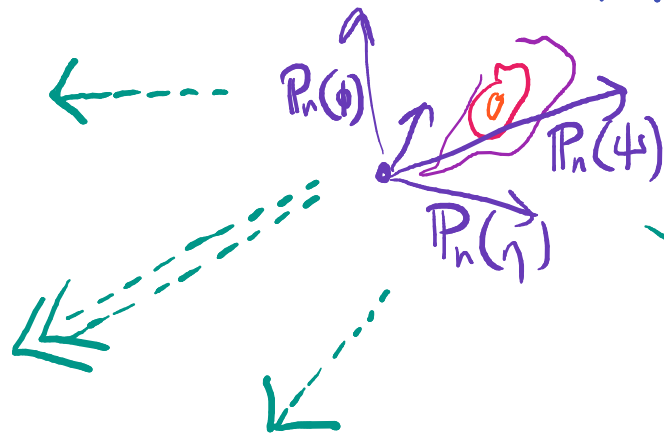
→ This requires the distribution over large sentences $P_{n-1}(Q_n = Q_n)$ to marginalize out to the small beliefs $P_{n-1}(\phi)$; however, $P_{n-1}(Q_n = Q_n)$ may disrespect many other small beliefs about future beliefs, EG a belief that $P_n(\phi)$ and $P_n(\psi)$ are correlated.

We might want things such as:

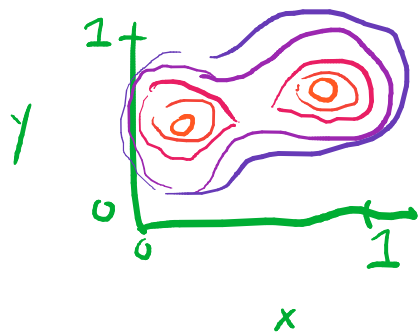
$$P_{n-1}(P_n(\phi)=x)$$



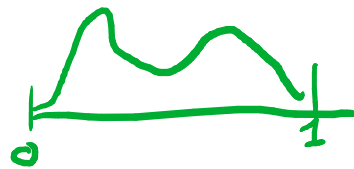
$P_{n-1}(Q_n=Q_n)$, our big
high-dimensional
distribution



$$P_{n-1}(P_n(\phi)=x \wedge P_n(\eta)=y)$$



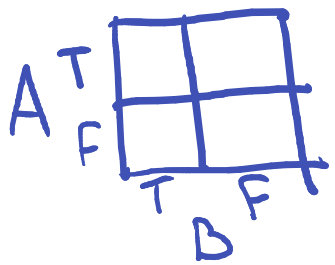
$$P_{n-1}(P_n(\eta)=x)$$



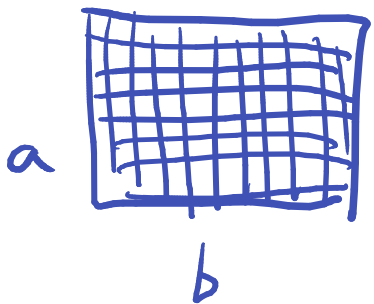
$$P_{n-1}(P_n(\phi) > .2)$$

$$E_{n-1}\left(\frac{P_n(\phi)^2 + P_n(\psi)^3}{P(\eta)}\right) > .2$$

Things like this are only possible if small beliefs place
consistent demands on the future.



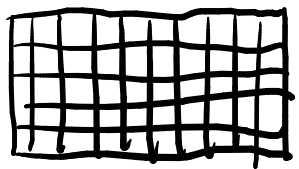
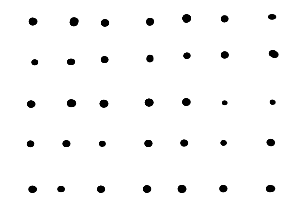
$A, B \in \{T, F\}$



$a, b \in \mathbb{R}$

Similarly to how we can enforce propositional coherence on a finite set of propositions S by enumerating the $2^{|S|}$ "worlds" & assigning probability to each, we can imagine ϵ -discretized real-coherence as dividing up all the real-valued variables mentioned in S into $\frac{1}{\epsilon}$ slices & assigning prob. to each.

Similar to how we can imagine extending to infinite S by assuming all but finitely many $\phi \in S$ to be independent & 50-50, we might assume any real-valued variable not mentioned by small beliefs to be independent & uniform.



Roman has been considering a dot-grid, which I think makes sense for the specific case of $P_n(P_m(\phi) = x)$ (ie beliefs about beliefs)

because we can stipulate beliefs to be multiples of some ϵ_n which grows quickly enough to not threaten the LIC.

However, it does seem more elegant to consider real-coherence more generally^{*}; perhaps it is relevant to decision theory due to expected values.

Since we cannot force arbitrary mathematically defined numbers to a dot-grid, a grid of bins seems necessary.

*: any bounded-range real-valued variable

- It would be nice to achieve real-coherence without any discretization at all, of course.

- What does real-value coherence mean, exactly?

Here are some formal things we might ask for:

- A. Contradictions of real-valued logic, like $a < b$ & $b < a$, should have probability zero. (What axioms exactly?)
- B. Individual beliefs like $P(x > .2)$, $P(.1 \leq x \leq .5)$, etc are consistent with some overall dist. on x Must be decidable!
- C. Similarly, multi-variable beliefs like $P(x > y)$ are consistent with some joint distribution
- D. Joint distributions on multiple variables } Some versions of this follow directly from propositional coherence!
marginalize correctly: $\int p(x, y) dx = p(y)$
- E. Distributions on mathematically related random variables are appropriately related; eg, if $z = x + y^2 + 2$, then $\mathbb{E}(z) = \mathbb{E}(x) + \mathbb{E}(y^2) + 2$ and so on

Does real-value coherence simplify BLI?

1. $P_n(\phi) \approx Q_n(\phi)$ for small ϕ

2. $P_n(\phi | Q_m = Q_m) = Q_m[\phi]$ $m > n$

3. $P_n(\phi | Q_m = Q_m \wedge Q_{m+1} = Q_{m+1}) = Q_{m+1}[\phi]$

4. $P_{n+1}(\phi) = \sum_Q P_{n+1}(Q_n = Q_n) Q_n(\phi)$

This arguably rolls 2 things into 1:

↳ Coherence between $P_n(Q_m = Q_m)$ and its marginals, $P_n(Q_m(\phi) = x)$

↳ faith in those marginals, $P_n(\phi | Q_m(\phi) = x) = x$